Le Chatelier's principle

Push a chemical equilibrium and it pushes back – but only partly. That is Le Chatelier's principle in one sentence. It is one of chemistry's most useful rules of thumb, and also one of the most misunderstood. This page works through four examples, shows them in interactive simulations, and ends with the case where the principle, as usually stated in words, actually gives the wrong answer.

The principle

The French chemist Henri Louis Le Chatelier (1850–1936) stated the principle in 1884, and the German physicist Ferdinand Braun gave it a more general form in 1887. It is therefore also called the Le Chatelier–Braun principle. In its school form it reads:

> **If a system at equilibrium is subjected to an external change, the equilibrium shifts in the direction that partly counteracts the change.**

The three classic changes are a change of *concentration*, of *pressure* (by changing the volume) and of *temperature*. A catalyst does not shift the equilibrium – it only makes it establish itself faster, because it lowers the activation energy equally in both directions.

The important word is *partly*. The equilibrium can never cancel the change completely. If you add a substance, there is always more of it in the end than before – just not as much more as you added.

Why it works: Q and K

The principle is not an independent law of nature but a consequence of thermodynamics. For a reaction $a\,\mathrm{A} + b\,\mathrm{B} \rightleftharpoons c\,\mathrm{C} + d\,\mathrm{D}$ we define the reaction quotient

$$Q = \frac{[\mathrm{C}]^{c}\,[\mathrm{D}]^{d}}{[\mathrm{A}]^{a}\,[\mathrm{B}]^{b}}$$

At equilibrium Q equals the equilibrium constant K. The link to the Gibbs energy is

$$\Delta_r G = RT \ln \frac{Q}{K}$$

If $Q < K$, then $\Delta_r G < 0$ and the reaction runs to the right. If $Q > K$, it runs to the left. That is the whole mechanism: a change alters either Q (concentration and pressure) or K (temperature), and the system moves until the two are equal again.

What changes Q?

Adding or removing a substance, and changing the volume or pressure. K is unchanged, so the equilibrium moves until $Q = K$ again.

What changes K?

Only temperature. The van 't Hoff equation $\dfrac{d\ln K}{dT} = \dfrac{\Delta_r H^\circ}{RT^2}$ shows that K increases with T for endothermic reactions and decreases for exothermic ones.

The best advice for students is therefore: **when in doubt, calculate Q and compare it with K.** That always gives the right answer – also in the cases where the principle stated in words is ambiguous.

Example 1 · Concentration: $H_2 + I_2 \rightleftharpoons 2HI$

At about 450 °C, $K = 50$ for the formation of hydrogen iodide. We start at equilibrium with $[\mathrm{H_2}] = [\mathrm{I_2}] = 0.100$ M. Then

$$[\mathrm{HI}] = \sqrt{K}\cdot 0.100\ \mathrm{M} = 0.707\ \mathrm{M}$$

Now 0.100 M H_2 is added, so $[\mathrm{H_2}] = 0.200$ M. Just after the addition

$$Q = \frac{0.707^2}{0.200\cdot 0.100} = 25 < K = 50$$

The equilibrium shifts to the right. Calling the amount of H_2 that reacts x, we need

$$\frac{(0.707 + 2x)^2}{(0.200 - x)(0.100 - x)} = 50 \quad\Longrightarrow\quad 46x^2 - 17.83x + 0.50 = 0$$

The physically possible root is $x = 0.0304$ M. The new equilibrium is

	$[H_2]$ / M	$[I_2]$ / M	[HI] / M	Q
Before	0.100	0.100	0.707	50
Just after addition	0.200	0.100	0.707	25
New equilibrium	0.170	0.070	0.768	50

Only about 30 % of the added H_2 is used up. The equilibrium counteracts the change, but only partly: there is still 70 % more H_2 than before.

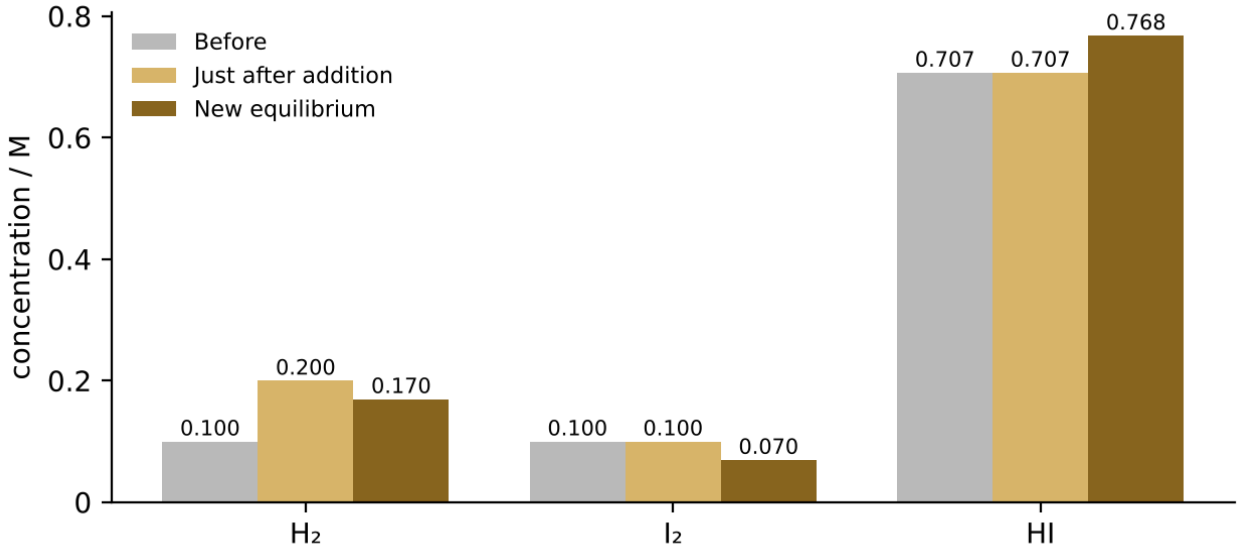


Figure 1. The concentrations before the addition, just after it, and at the new equilibrium.

Example 2 · Pressure: $N_2O_4 \rightleftharpoons 2NO_2$

Dinitrogen tetroxide is colourless; nitrogen dioxide is red-brown. At 25 °C, $K_p = 0.15$ (with the standard pressure $p^\circ = 1$ bar). Starting from pure N_2O_4 and letting a fraction α dissociate, the mole fractions are $x_{NO_2} = \frac{2\alpha}{1+\alpha}$ and $x_{N_2O_4} = \frac{1-\alpha}{1+\alpha}$. This gives

$$K_p = \frac{x_{NO_2}^2}{x_{N_2O_4}} \cdot \frac{P}{p^\circ} = \frac{4\alpha^2}{1-\alpha^2} \cdot \frac{P}{p^\circ} \quad \Longrightarrow \quad \alpha = \sqrt{\frac{K_p}{K_p + 4P/p^\circ}}$$

P / bar	α	x(NO_2)
1	0.190	0.32
10	0.061	0.12

Increasing the pressure tenfold lowers the degree of dissociation from 19 % to 6 %. The equilibrium shifts towards the side with fewer gas molecules. But notice what you actually *see* when you push in the plunger of a syringe: the gas first becomes *darker*, because the same NO_2

molecules are now squeezed into a smaller space. Then it slowly becomes somewhat lighter again as the equilibrium shifts towards N_2O_4. It never becomes as light as before, though – the NO_2 concentration still ends up higher. Again: partial counteraction.

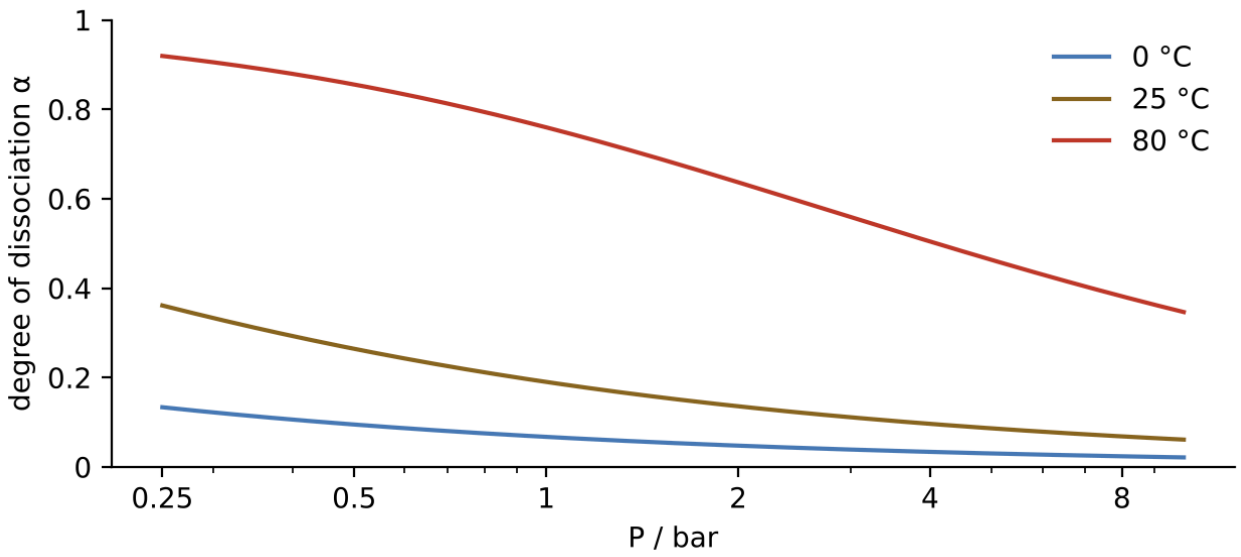


Figure 2. The degree of dissociation α for $N_2O_4 \rightleftharpoons 2NO_2$ as a function of pressure at three temperatures.

Example 3 · Temperature and van 't Hoff

The dissociation of N_2O_4 is endothermic with $\Delta_r H^\circ \approx +57$ kJ/mol. Assuming $\Delta_r H^\circ$ is constant, integrating the van 't Hoff equation gives

$$\ln\frac{K_2}{K_1} = -\frac{\Delta_r H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

From 298 K to 350 K:

$$\ln\frac{K_2}{K_1} = -\frac{57\,200}{8.314}\left(\frac{1}{350} - \frac{1}{298}\right) = 3.43 \quad\Longrightarrow\quad K_2 = 0.15 \cdot e^{3.43} \approx 4.6$$

At 1 bar this gives $\alpha^2 = 4.6/(4.6+4)$, so $\alpha \approx 0.73$. The dissociation rises from 19 % to 73 %, and the brown colour deepens visibly. This is the classic demonstration with two sealed tubes of NO_2, one in hot water and one in ice water. Use the "Hot water bath" and "Ice water" buttons in the simulation above.

Temperature is the only one of the three changes that alters K itself. When you heat the system, it "counteracts" the change by letting the heat-absorbing reaction run.

Application: the Haber–Bosch compromise

The ammonia synthesis $\mathbf{N_2 + 3H_2 \rightleftharpoons 2NH_3}$ is exothermic ($\Delta_r H^\circ \approx -92$ kJ/mol) and reduces the number of gas molecules from four to two. Le Chatelier's principle therefore says: *low temperature and high pressure* give the highest yield. But at low temperature the reaction is far too slow, even with an iron catalyst. Industry chooses a compromise around 400–500 °C and 150–300 bar, and continuously removes ammonia by condensing it out as a liquid, so that the equilibrium is constantly pulled to the right.

The curves are calculated for an N_2/H_2 mixture in the ratio 1 : 3, ideal gas, $K_p(723\ \mathrm{K}) \approx 1.45 \cdot 10^{-5}$ bar^{-2} and constant $\Delta_r H^\circ$. At very high pressures real gases deviate from the ideal gas law, and the measured yields are somewhat higher than the curves.

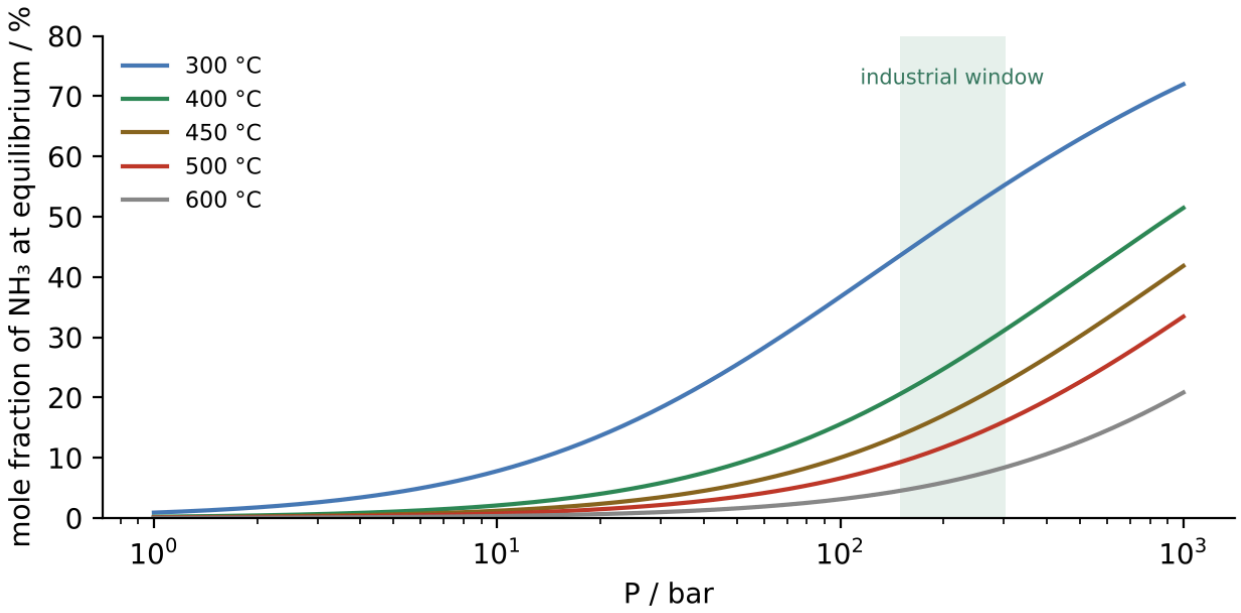


Figure 3. Mole fraction of NH_3 at equilibrium as a function of pressure at five temperatures.

Example 4 · The exception: more nitrogen, less ammonia

Here is a question almost everyone gets wrong: *An ammonia equilibrium is held at constant pressure and temperature. What happens when more N_2 is added?* The principle in words says: the equilibrium shifts to the right to use up some of the added N_2. But that is not always true.

At constant total pressure P, the reaction quotient can be written in terms of amounts:

$$Q = \frac{x_{\mathrm{NH_3}}^2}{x_{\mathrm{N_2}}\, x_{\mathrm{H_2}}^3}\left(\frac{p^\circ}{P}\right)^2 = \frac{n_{\mathrm{NH_3}}^2\, n_{\mathrm{tot}}^2}{n_{\mathrm{N_2}}\, n_{\mathrm{H_2}}^3}\left(\frac{p^\circ}{P}\right)^2$$

Adding N_2 increases both $n_{\mathrm{N_2}}$ in the denominator and n_{tot} in the numerator. Differentiating gives

$$\frac{\partial \ln Q}{\partial n_{\mathrm{N_2}}} = \frac{2}{n_{\mathrm{tot}}} - \frac{1}{n_{\mathrm{N_2}}}$$

This is positive precisely when $x_{\mathrm{N_2}} > \frac{1}{2}$. In a nitrogen-rich mixture, added N_2 therefore *increases* Q, so $Q > K$, and the equilibrium shifts to the left: ammonia decomposes and even more N_2 is formed.

A simple numerical example

Take $n_{\mathrm{N_2}} = 6$, $n_{\mathrm{H_2}} = 1$ and $n_{\mathrm{NH_3}} = 1$ mol, so $n_{\mathrm{tot}} = 8$ and $x_{\mathrm{N_2}} = 0.75$. Apart from the common factor $(p^\circ/P)^2$, $Q \propto 1 \cdot 64/(6 \cdot 1) = 10.7$. Adding 1 mol N_2 gives $n_{\mathrm{N_2}} = 7$ and $n_{\mathrm{tot}} = 9$, and $Q \propto 81/7 = 11.6$. The reaction quotient has increased.

With the real equilibrium constant

At 450 °C and 300 bar, a mixture made from 6 mol N_2 and 1 mol H_2 settles at 5.889 mol N_2, 0.666 mol H_2 and 0.222 mol NH_3 ($x_{\mathrm{N_2}} = 0.87$). Adding 0.5 mol N_2 at fixed pressure and temperature leaves 0.219 mol NH_3 – *less* than before – and 6.391 mol N_2, slightly *more* than the 6.389 mol present just after the addition. With a stoichiometric mixture (1 : 3), by contrast, things go as the principle says.

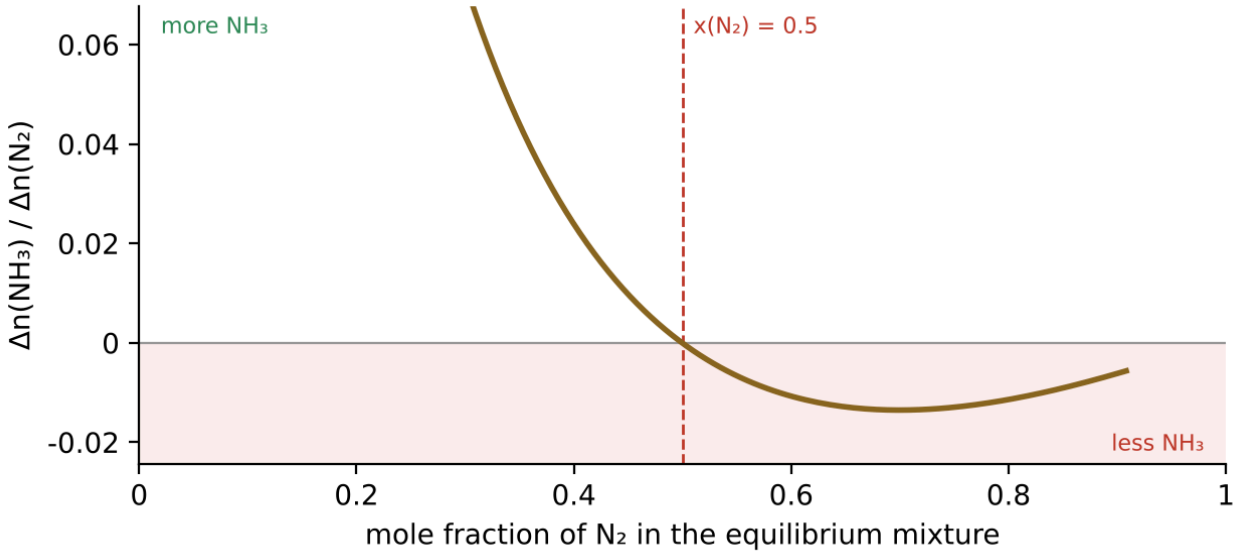


Figure 4. The change in the amount of NH_3 when N_2 is added at constant P and T, as a function of the mole fraction of N_2 in the equilibrium mixture. The sign changes at $x(N_2) = 0.5$.

The point is not that Le Chatelier was wrong, but that "the change" is ambiguous. If N_2 is added at constant *volume*, its concentration rises and the equilibrium always moves to the right. At constant *pressure* the other substances are diluted at the same time, and that effect can win. Paul Ehrenfest pointed out as early as 1911 that the principle in its usual form does not say precisely what is held constant. Thermodynamics has no such ambiguity: Q compared with K settles the matter.

Inert gas is a related case. Adding argon at constant volume raises the total pressure, but the partial pressures are unchanged and the equilibrium does not move at all. Adding argon at constant pressure amounts to a dilution, and the equilibrium shifts towards the side with more gas molecules.

In everyday life

Fizzy drinks

$\mathrm{CO_2(g) \rightleftharpoons CO_2(aq)}$. When the bottle is opened, the pressure above the liquid drops and the equilibrium shifts towards the gas phase – it fizzes. A warm drink fizzes more, because dissolving CO_2 is exothermic.

Blood and mountains

$\mathrm{Hb + 4\,O_2 \rightleftharpoons Hb(O_2)_4}$. In thin mountain air the partial pressure of O_2 is low, so less haemoglobin is oxygenated. Over weeks, the body responds by making more red blood cells.

Ocean acidification

$\mathrm{CO_2 + H_2O \rightleftharpoons H^+ + HCO_3^-}$. More CO_2 in the atmosphere pushes the equilibrium to the right, the pH falls, and H^+ reacts with carbonate, which shellfish and corals use for their shells.

Ice under pressure

Ice takes up more room than water. Increasing the pressure shifts the equilibrium ice ⇌ water towards the denser phase, so the melting point falls slightly – about 0.0074 °C per bar.

History

Henri Le Chatelier was an engineer and chemist at the École des Mines in Paris and worked on cement, metallurgy and high-temperature measurement. Among other things he invented a thermocouple of platinum and platinum–rhodium. His principle of 1884 built on J.H. van 't Hoff's

work on the effect of temperature on equilibrium, and Ferdinand Braun – later known for the cathode-ray tube and the Nobel Prize for wireless telegraphy – generalised it in 1887.

Ironically, around 1901 Le Chatelier himself tried to synthesise ammonia at high pressure, exactly as his principle prescribed. The attempt ended in an explosion because air had got into the apparatus, and he gave up. Fritz Haber succeeded in 1909, and Carl Bosch scaled the process up at BASF in 1913. Le Chatelier is said to have later regretted letting the ammonia synthesis slip through his fingers.

Exercises

1. Add I_2 instead of H_2

Start again from $[\mathrm{H_2}] = [\mathrm{I_2}] = 0.100$ M and $[\mathrm{HI}] = 0.707$ M at $K = 50$, and add 0.100 M I_2. Find the new equilibrium.

Answer: By symmetry the result is the same as in Example 1 with H_2 and I_2 swapped: $[\mathrm{H_2}] = 0.070$ M, $[\mathrm{I_2}] = 0.170$ M and $[\mathrm{HI}] = 0.768$ M.

2. Halve the volume of the HI equilibrium

What happens to the equilibrium $H_2 + I_2 \rightleftharpoons 2HI$ when the volume is halved?

Answer: All concentrations double. Since there are two molecules on each side, the factor $2^2/(2 \cdot 2) = 1$ cancels and $Q = K$ still. The equilibrium does not move, even though the pressure doubles.

3. N_2O_4 at 0 °C

Use $K_p(298\ \mathrm{K}) = 0.15$ and $\Delta_r H^\circ = 57.2$ kJ/mol to find K_p and α at 273 K and 1 bar.

Answer: $\ln(K_2/K_1) = -\frac{57\,200}{8.314}\left(\frac{1}{273} - \frac{1}{298}\right) = -2.11$, so $K_2 \approx 0.018$. This gives $\alpha = \sqrt{0.018/4.018} \approx 0.067$. The gas is almost colourless.

4. When does the sign change?

Show that adding H_2 at constant pressure and temperature always shifts the ammonia equilibrium to the right.

Answer: $\partial \ln Q / \partial n_{\mathrm{H_2}} = 2/n_{\mathrm{tot}} - 3/n_{\mathrm{H_2}}$. For this to be positive would require $x_{\mathrm{H_2}} > \frac{3}{2}$, which is impossible. Q always falls and the equilibrium moves to the right.

5. A general rule

Find a general condition for when adding a substance i at constant P and T shifts the equilibrium towards the side where i appears – that is, "the wrong way".

Answer: With stoichiometric coefficients ν_j (positive for products, negative for reactants) and $\Delta\nu = \sum_j \nu_j$, $\ln Q = \sum_j \nu_j \ln n_j - \Delta\nu \ln n_{\text{tot}} + \text{const.}$ Hence $\partial \ln Q / \partial n_i = \nu_i/n_i - \Delta\nu/n_{\text{tot}}$. For a reactant ($\nu_i < 0$) in a reaction with $\Delta\nu < 0$, this becomes positive when $x_i > \nu_i/\Delta\nu$. For N_2 in the ammonia synthesis, $\nu_i/\Delta\nu = (-1)/(-2) = \tfrac{1}{2}$.

See also the Gibbs paradox page.

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